

A gcd-sum evaluation of a signed square-part function: a proof of Schulte’s conjecture on OEIS A358272

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Abstract

Let λ denote the Liouville function and let a be the multiplicative function determined by $a(p^e) = (-1)^e p^{2\lfloor e/2 \rfloor}$. In November 2022 Werner Schulte conjectured on OEIS A358272 that $a(n) = \sum_{k=1}^n \gcd(k, n) \lambda(\gcd(k, n))$ for all $n \geq 1$. We prove it. The argument is the classical gcd-sum rearrangement — group the k in $\{1, \dots, n\}$ by the value of $\gcd(k, n)$, which turns any sum of that shape into a Dirichlet convolution with Euler’s totient — followed by a two-line evaluation of the resulting local factor at a prime power, where an alternating sum collapses according to the parity of the exponent. The result is elementary, and we say so; its interest is that the statement has stood labelled as a conjecture on the entry since 2022.

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1 The sequence and the conjecture

Throughout, p denotes a prime, φ is Euler’s totient function (OEIS A000010) and λ is the Liouville function (OEIS A008836), so that $\lambda(n) = (-1)^{\Omega(n)}$ where $\Omega(n)$ counts the prime factors of n with multiplicity. Both are multiplicative, and λ is completely multiplicative.

OEIS A358272 is defined as the multiplicative function

$$a(p^e) = (-1)^e p^{2\lfloor e/2 \rfloor} \quad (p \text{ prime, } e \geq 0), \tag{1}$$

extended to all $n \geq 1$ by $a(n) = \prod_{p \parallel n} a(p^e)$, with $a(1) = 1$. The entry has offset 1 and begins

$$a(1), \dots, a(12) = 1, -1, -1, 4, -1, 1, -1, -4, 9, 1, -1, -4.$$

Since $p^{2\lfloor e/2 \rfloor}$ is the largest square dividing p^e , the function $|a|$ is the square part of n (OEIS A008833), and $a = \lambda \cdot A008833$; the entry describes itself as the signed version of A008833.

The entry carries the following line in its FORMULA section:

Conjecture: $a(n) = \text{Sum}_{k=1..n} \gcd(k, n) * \text{lambda}(\gcd(k, n))$ for $n > 0$.

The conjecture is due to Werner Schulte, who created the entry on 9 November 2022. As of the “Last modified” line on the live entry (15 November 2025, revision 23) the statement is still recorded as a conjecture; the only link on the entry is a table of values by Andrew Howroyd, and no proof appears among the entry’s links, formulas or programs.

Write

$$S(n) = \sum_{k=1}^n \gcd(k, n) \lambda(\gcd(k, n)). \tag{2}$$

The claim to be proved is that $S(n) = a(n)$ for every $n \geq 1$.

2 The gcd-sum lemma

Sums of the shape (2) depend on k only through $\gcd(k, n)$, so they can be collapsed by grouping the summation range according to that value. This is the standard device behind Pillai's arithmetical function $\sum_{k \leq n} \gcd(k, n)$, and it is all the arithmetic input the theorem needs.

Lemma 1. *Let f be any arithmetic function and let $n \geq 1$. Then*

$$\sum_{k=1}^n f(\gcd(k, n)) = \sum_{d|n} f(d) \varphi(n/d) = (f * \varphi)(n),$$

where $*$ denotes Dirichlet convolution.

Proof. Every k with $1 \leq k \leq n$ satisfies $\gcd(k, n) = d$ for exactly one divisor d of n , so the summation range splits as a disjoint union over the divisors of n :

$$\sum_{k=1}^n f(\gcd(k, n)) = \sum_{d|n} f(d) \cdot \#\{k : 1 \leq k \leq n, \gcd(k, n) = d\}.$$

Fix such a d and write $k = dm$. The condition $1 \leq k \leq n$ becomes $1 \leq m \leq n/d$, and $\gcd(k, n) = d$ becomes $\gcd(m, n/d) = 1$. Hence the inner count is the number of integers m in $[1, n/d]$ coprime to n/d , which is $\varphi(n/d)$ by definition. Substituting gives the middle expression, and since $d \mapsto n/d$ runs over the divisors of n as d does, that expression is exactly the Dirichlet convolution $(f * \varphi)(n)$. $\square$

Applying Lemma 1 to (2) with $f(d) = d \lambda(d)$ gives

$$S = f * \varphi, \quad f(d) = d \lambda(d). \quad (3)$$

Both f and φ are multiplicative — f because it is a product of the completely multiplicative functions $d \mapsto d$ and λ — and the Dirichlet convolution of two multiplicative functions is multiplicative. So S is multiplicative, and to prove $S = a$ it suffices to compare the two at a single prime power.

3 The theorem

Theorem 2. *For every $n \geq 1$,*

$$\sum_{k=1}^n \gcd(k, n) \lambda(\gcd(k, n)) = a(n),$$

where a is the multiplicative function defined by (1). That is, Schulte's conjecture on OEIS A358272 is true.

Proof. By (3) the function S is multiplicative, as is a , so it is enough to show $S(p^e) = a(p^e)$ for every prime p and every $e \geq 0$. For $e = 0$ both sides equal 1, so assume $e \geq 1$.

With $f(d) = d \lambda(d)$ we have $f(p^j) = (-1)^j p^j$, and $\varphi(p^i) = p^i - p^{i-1}$ for $i \geq 1$ while $\varphi(1) = 1$. Separating the top term $j = e$ of the convolution,

$$S(p^e) = \sum_{j=0}^e f(p^j) \varphi(p^{e-j}) = (-1)^e p^e + \sum_{j=0}^{e-1} (-1)^j p^j (p^{e-j} - p^{e-j-1}).$$

In the remaining sum the powers of p no longer depend on j : indeed $p^j(p^{e-j} - p^{e-j-1}) = p^e - p^{e-1}$ for every j in the range. Therefore

$$S(p^e) = (-1)^e p^e + (p^e - p^{e-1}) \sum_{j=0}^{e-1} (-1)^j, \quad (4)$$

and the whole computation has been reduced to an alternating sum of ± 1 , which is 0 when e is even and 1 when e is odd.

If e is even, (4) gives $S(p^e) = p^e$, while $a(p^e) = (-1)^e p^{2\lfloor e/2 \rfloor} = p^e$ because $2\lfloor e/2 \rfloor = e$.

If e is odd, (4) gives

$$S(p^e) = -p^e + (p^e - p^{e-1}) = -p^{e-1},$$

while $a(p^e) = (-1)^e p^{2\lfloor e/2 \rfloor} = -p^{e-1}$ because $2\lfloor e/2 \rfloor = e - 1$.

In both cases $S(p^e) = a(p^e)$, and multiplicativity finishes the proof. $\square$

Remark 3. The proof explains the shape of (1) rather than merely confirming it. The factor $(-1)^e$ and the floor in the exponent are not two independent features: both come from the single alternating sum in (4). The sum vanishes exactly when e is even, which is precisely when the top term $(-1)^e p^e$ survives intact; when e is odd the top term is cancelled and what remains is the single leftover $-p^{e-1}$.

4 Verification

The identity was checked numerically by two deliberately different routes, neither of which was used in the proof.

The first computes the right-hand side of (2) directly, forming $\gcd(k, n)$ for each $k = 1, \dots, n$ and evaluating λ on it by factorisation. The second builds $a(n)$ from the multiplicative definition (1) alone, factoring n and taking the product of the local values; it never forms a gcd. The two agree with each other for all n with $1 \leq n \leq 400$, and each agrees with all 80 terms of the DATA section published on the entry, over its full range $n = 1, \dots, 80$.

5 Concluding remarks

The result is elementary, and the entry itself stands close to it. On 11 May 2025 Ridouane Oudra added to A358272 the unlabelled formula

$$a(n) = \sum_{d|n} \lambda(d) d \varphi(n/d),$$

which is the statement $a = f * \varphi$ of (3). Given that line, the conjecture is exactly one application of Lemma 1 away, and the content of this note is the observation that the two records on the entry — one labelled a conjecture, one stated as fact — are the same identity. What Theorem 2 adds is a proof that does not presuppose either.

The companion entry OEIS A358319, also due to Schulte, carries the conjecture $b(n) = \sum_{k=1}^n \gcd(k, n) \mu(\text{rad}(\gcd(k, n)))$ for the multiplicative function with $b(p^e) = ((p-2) - (p-1)e)p^{e-1}$. It yields to Lemma 1 in the same way, with a different local computation; that case is treated separately.

Finally, a nearby conjecture of the same family is already settled and should not be confused with these. OEIS A384531 carries Schulte's conjecture $c(n) = \sum_{i=1}^n \gcd(i, n) \gcd(i+1, n)$; the entry links a note of David G. Radcliffe of June 2025 which proves it, by a direct proof of multiplicativity together with the identity $c(p^e) = 2A018804(p^e) - p^e$. That statement is therefore not open, notwithstanding the label still carried on the entry.

References

- [1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, <https://oeis.org>, 2026. Sequence A358272, formula of W. Schulte, 9 November 2022, and formula of R. Oudra, 11 May 2025; sequences A000010, A008833, A008836, A358319.
- [2] D. G. Radcliffe, *Notes on two arithmetical functions*, June 2025; linked from OEIS A384531.